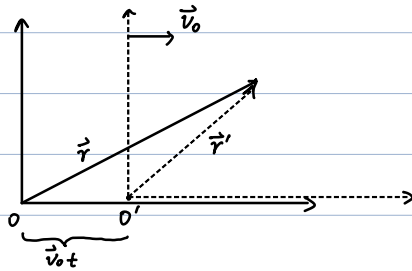


Conservation of Momentum

Revisit work - kinetic energy theorem

$$K_f - K_i = W$$

Invariant under Galilean transformation:



$$\begin{aligned}\vec{r}' &= \vec{r} - \vec{v}_0 t \\ \Rightarrow \begin{cases} \vec{v}' &= \vec{v} - \vec{v}_0 \\ \vec{a}' &= \vec{a} \end{cases}\end{aligned}$$

Assume 1D and constant force

$$\begin{cases} W = Fd \\ W' = Fd' = F(d - v_0 t) \end{cases}$$

$$K' = \frac{1}{2} m (v')^2 = \frac{1}{2} m (v - v_0)^2 = K - v_0 m v + \frac{1}{2} m v_0^2$$

$$K'_f - K'_i = K_f - K_i - v_0 m (v_f - v_i)$$

$$= Fd - v_0 m (v_f - v_i)$$

$$= \underbrace{F(d - v_0 t)}_{d'} + Fv_0 t - v_0 m (v_f - v_i)$$

$$= W' + v_0 [Ft - m(v_f - v_i)]$$

$\Rightarrow$ as long as

$$Ft = m(v_f - v_i) \quad \rightarrow \text{obviously true: } v_f = v_i + at \quad (\text{constant acceleration})$$

then

$$K'_f - K'_i = W'$$

i.e. work - kinetic energy theorem always holds

(in different inertial frames)

Define

linear momentum : $\vec{p} \equiv m \vec{v}$

(often just called "momentum")

Impulse : $\vec{I} \equiv \int_{t_i}^{t_f} \vec{F} dt$

Impulse - Momentum Theorem:

$$\vec{I} = \vec{p}_f - \vec{p}_i$$

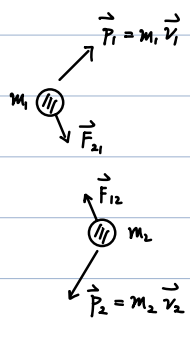
proof:

$$\begin{aligned}\vec{F} &= m \frac{d\vec{v}}{dt} \\ \Rightarrow \boxed{\vec{F} = \frac{d\vec{p}}{dt}} & \quad \text{differential form} \\ \Rightarrow \int_{t_i}^{t_f} \vec{F} dt &= \int_{\vec{p}_i}^{\vec{p}_f} d\vec{p} \\ \Rightarrow I &= \vec{p}_f - \vec{p}_i\end{aligned}$$

e.g. constant force

$$\vec{F}t = m(\vec{v}_f - \vec{v}_i) \longrightarrow \text{(we just used)}$$

Isolated Two-particle system


$$\begin{aligned}\left\{ \begin{array}{l} \vec{F}_{21} = \frac{d\vec{p}_1}{dt}, \quad \vec{F}_{12} = \frac{d\vec{p}_2}{dt} \\ \vec{F}_{21} = -\vec{F}_{12} \end{array} \right. \\ \Rightarrow \frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} = 0 \\ \Rightarrow \vec{p}_1 + \vec{p}_2 = \text{constant} \\ \vec{p}_{1i} + \vec{p}_{2i} = \vec{p}_{1f} + \vec{p}_{2f}\end{aligned}$$

Generalization : Conservation of Linear Momentum

$$\vec{P} = \sum_i \vec{p}_i = \text{constant}$$

if the system is $\left\{ \begin{array}{l} \text{closed (no particles leave or enter)} \\ \text{isolated (no external force)} \end{array} \right.$

(2)

Origin of Conservation Laws: Symmetry

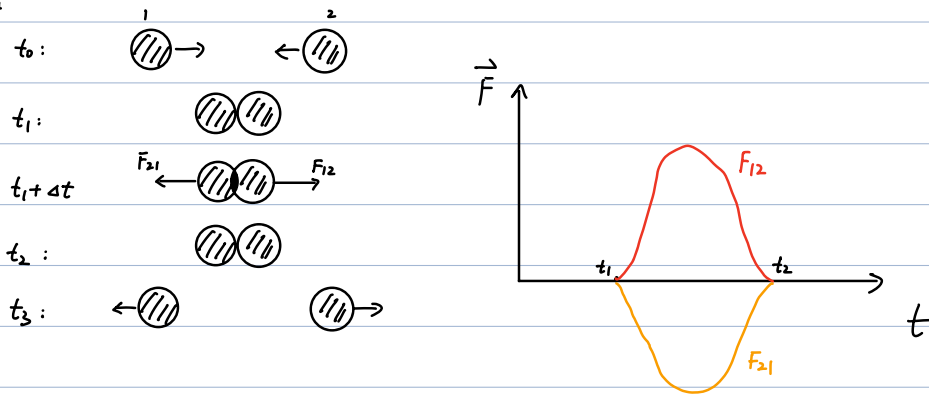
- ① Energy : time translation invariance
- ② Linear Momentum : spatial translation invariance
- ...

Proof be show in Theoretical Mechanics (later course in physics)

Practical use:

- sanity check
- applicable without knowing $\vec{F}$
- convenience in problem solving.

• Collision



$$\int_{t_1}^{t_2} F_{12} dt = - \int_{t_1}^{t_2} F_{21} dt$$

$$\Rightarrow p_2^f - p_2^i = - (p_1^f - p_1^i)$$

$$\text{i.e. } p_1^i + p_2^i = p_1^f + p_2^f$$

conservation of total linear momentum $\vec{P} \equiv \vec{P}_1 + \vec{P}_2$

(again, requires closed, isolated system)

• Kinetic energy NOT necessarily conserved in collisions!

- Elastic collision : $K_i = K_f$
- Inelastic collision : $K_i > K_f$

- Perfectly inelastic:

$$\vec{v}_{1f} = \vec{v}_{2f} \equiv \vec{v}_f$$

Momentum conservation:

$$m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i} = m_1 \vec{v}_f + m_2 \vec{v}_f$$

$$\Rightarrow \vec{v}_f = \frac{m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i}}{m_1 + m_2}$$

Loss of kinetic energy:

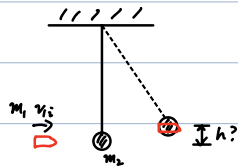
$$K_i - K_f = \frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 - \frac{1}{2} (m_1 + m_2) v_f^2$$

$$= \frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2$$

$$- \frac{1}{2(m_1 + m_2)} (m_1^2 v_{1i}^2 + m_2^2 v_{2i}^2 + 2m_1 m_2 \vec{v}_{1i} \cdot \vec{v}_{2i})$$

$$= \frac{m_1 m_2}{2(m_1 + m_2)} (\vec{v}_{1i} - \vec{v}_{2i})^2 \quad \begin{cases} > 0 & \vec{v}_{1i} \neq \vec{v}_{2i} \\ = 0 & \vec{v}_{1i} = \vec{v}_{2i} \quad (\text{no collision}) \end{cases}$$

Example.

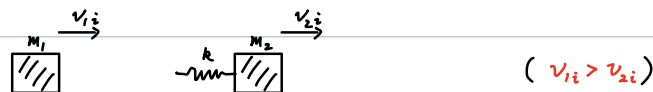


$$v_f = \frac{m_1 v_{1i}}{m_1 + m_2} \quad (\text{momentum conservation})$$

$$(m_1 + m_2) g h = \frac{1}{2} (m_1 + m_2) v_f^2$$

$$\Rightarrow h = \frac{1}{2g} v_f^2 = \frac{m_1^2 v_{1i}^2}{2(m_1 + m_2)^2 g}$$

Example.



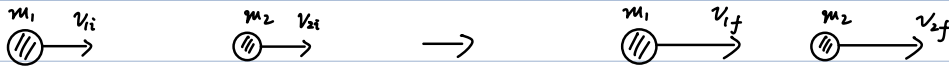
Q: maximal deformation " x " of spring?

Note: "maximal" $\leftrightarrow$ perfectly inelastic

$$\frac{m_1 m_2}{2(m_1 + m_2)} (v_{1i} - v_{2i})^2 = \frac{1}{2} k x^2$$

$$\Rightarrow x = \sqrt{\frac{m_1 m_2}{k(m_1 + m_2)}} (v_{1i} - v_{2i})$$

- Elastic collision in 1D (head to head)



$$\begin{cases} \frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 & (\text{energy}) \\ m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} & (\text{momentum}) \end{cases}$$

$$\Rightarrow \begin{cases} v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} + \frac{2m_2}{m_1 + m_2} v_{2i} \\ v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i} + \frac{m_2 - m_1}{m_1 + m_2} v_{2i} \end{cases}$$

Special cases:

① $v_{2i} = 0$

$$\Rightarrow \begin{cases} v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} \\ v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i} \end{cases}$$

if $m_1 \ll m_2 \quad \Rightarrow \quad \text{⊙}$

$$\Rightarrow \begin{cases} v_{1f} \approx -v_{1i} \\ v_{2f} \approx 0 \end{cases}$$

if $m_1 \gg m_2 \quad \text{⊙} \rightarrow \text{○}$

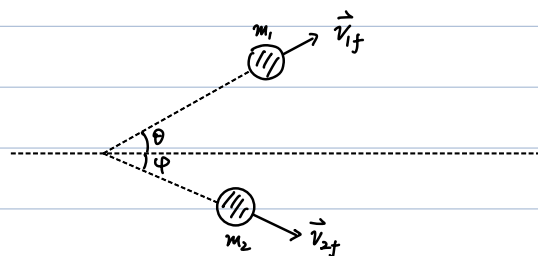
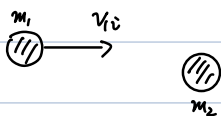
$$\Rightarrow \begin{cases} v_{1f} \approx v_{1i} \\ v_{2f} \approx 2v_{1i} \end{cases}$$

(think: easier to visualize in the frame of v_{1i})

② $m_1 = m_2$

$$\Rightarrow \begin{cases} v_{1f} = v_{2i} \\ v_{2f} = v_{1i} \end{cases} \quad (\text{exchange of velocity})$$

- Elastic collision in 2D



⑤

$$\left\{ \begin{array}{l} \frac{1}{2} m_1 v_{1i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 \quad (\text{energy}) \\ m_1 v_{1i} = m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \varphi \quad (\text{momentum}) \\ 0 = m_1 v_{1f} \sin \theta - m_2 v_{2f} \sin \varphi \end{array} \right.$$

3 eqs. 4 vars $\{v_{1f}, v_{2f}, \theta, \varphi\}$

Need one more equation to solve (e.g. given value of θ)

special case: $m_1 = m_2$

$$\left\{ \begin{array}{l} v_{1i}^2 = v_{1f}^2 + v_{2f}^2 \\ \vec{v}_{1i} = \vec{v}_{1f} + \vec{v}_{2f} \end{array} \right.$$

$$\Rightarrow (\vec{v}_{1f} + \vec{v}_{2f})^2 = v_{1f}^2 + v_{2f}^2$$

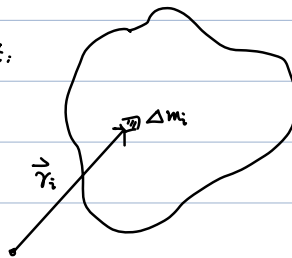
$$\Rightarrow \vec{v}_{1f} \cdot \vec{v}_{2f} = 0 \quad \text{always orthogonal}$$

Center of mass

2-particle: $\vec{r}_{cm} \equiv \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$

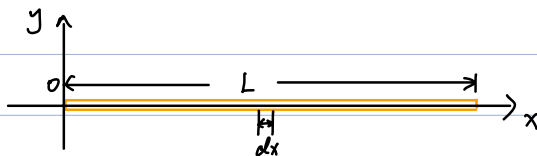
many-particle: $\vec{r}_{cm} \equiv \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$

extended object:



$$\begin{aligned} \vec{r}_{cm} &\equiv \lim_{\Delta m \rightarrow 0} \frac{\sum_i \Delta m_i \vec{r}_i}{M} \\ &= \frac{\int \vec{r} dm}{M} \\ &= \frac{\int \vec{r} \rho dV}{M} \end{aligned}$$

Example. Uniform Rod.



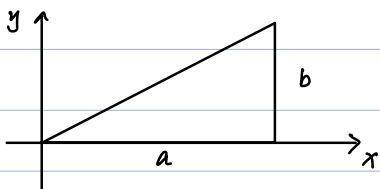
$$\begin{aligned} x_{cm} &= \frac{\int_0^L \lambda x dx}{\int_0^L \lambda dx} \\ &= \frac{\frac{1}{2} \lambda L^2}{\lambda L} \\ &= \frac{1}{2} L \end{aligned}$$

Example. Non-uniform Rod.

$$\lambda(x) = 2x$$

$$\begin{aligned} \Rightarrow x_{cm} &= \frac{\int_0^L \lambda(x) \cdot x \, dx}{\int_0^L \lambda(x) \, dx} \\ &= \frac{\int_0^L 2x^2 \, dx}{\int_0^L 2x \, dx} \\ &= \frac{\frac{2}{3} L^3}{\frac{1}{2} L^2} \\ &= \frac{2}{3} L \end{aligned}$$

Example. Uniform Triangle



$$\vec{r}_{cm} = \frac{\int \vec{r} \rho \, dx \, dy}{\int \rho \, dx \, dy}$$

$$\begin{aligned} x_{cm} &= \frac{\int_0^a x \, dx \int_0^{\frac{b}{a}x} \rho \, dy}{\int_0^a dx \int_0^{\frac{b}{a}x} \rho \, dy} \\ &= \frac{\int_0^a \frac{1}{2} x^2 \, dx}{\int_0^a \frac{b}{a} x \, dx} \\ &= \frac{\frac{1}{3} a^3}{\frac{1}{2} a^2} \\ &= \frac{2}{3} a \end{aligned}$$

$$\begin{aligned} y_{cm} &= \frac{\int_0^a dx \int_0^{\frac{b}{a}x} \rho y \, dy}{\int_0^a dx \int_0^{\frac{b}{a}x} \rho \, dy} \\ &= \frac{\int_0^a dx \cdot \frac{1}{2} \left(\frac{b}{a} x \right)^2}{\int_0^a \frac{b}{a} x \, dx} \\ &= \frac{\frac{1}{2} \frac{b^2}{a^2} \frac{1}{3} a^3}{\frac{b}{a} \frac{1}{2} a^2} \\ &= \frac{1}{3} b \end{aligned}$$

• Motion of a Many-particle system.

$$\vec{r}_{cm} = \frac{\sum_i m_i \vec{r}_i}{M}$$

$$\Rightarrow \vec{v}_{cm} \equiv \frac{d\vec{r}_{cm}}{dt} = \frac{\sum_i m_i \vec{v}_i}{M} = \frac{\sum_i \vec{p}_i}{M} \equiv \frac{\vec{p}_{tot}}{M}$$

$$\Rightarrow \vec{a}_{cm} \equiv \frac{d\vec{v}_{cm}}{dt} = \frac{\sum_i m_i \vec{a}_i}{M} = \frac{\sum_i \vec{F}_i}{M}$$

$$\Rightarrow \sum_i \vec{F}_i = M \vec{a}_{cm} = \frac{d\vec{p}_{tot}}{dt} \quad (\text{Impulse - Linear momentum})$$

Note: all internal forces cancel $\Rightarrow \sum_i \vec{F}_i = \sum_i \vec{F}_{ext}$
↓
external

- König Theorem - Kinetic energy in CM frame.

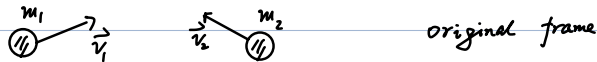
$$\begin{aligned}
 E_{ki} &= \frac{1}{2} m_i v_i^2 \\
 &= \frac{1}{2} m_i (\vec{v}_{cm} + \vec{v}_i') \cdot (\vec{v}_{cm} + \vec{v}_i') \\
 &= \frac{1}{2} m_i [v_{cm}^2 + (v_i')^2 + 2 \vec{v}_{cm} \cdot \vec{v}_i']
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow E_K &\equiv \sum_i E_{ki} \\
 &= \frac{1}{2} \sum_i m_i v_{cm}^2 + \frac{1}{2} \sum_i m_i (v_i')^2 + \vec{v}_{cm} \cdot \sum_i m_i \vec{v}_i' \\
 &= \frac{1}{2} M v_{cm}^2 + \frac{1}{2} \sum_i m_i (v_i')^2 + \vec{v}_{cm} \cdot \underbrace{\frac{d}{dt} \sum_i m_i \vec{r}_i'}_0 \\
 &= E_{K,cm} + E_K'
 \end{aligned}$$

Note: $\vec{r}_i' \equiv \vec{r}_i - \vec{r}_{cm}$

$$\begin{aligned}
 \vec{r}_{cm} &= \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i} \\
 \Rightarrow \sum_i m_i (\vec{r}_i - \vec{r}_{cm}) &= \sum_i m_i \vec{r}_i - \sum_i m_i \vec{r}_i = 0
 \end{aligned}$$

- Total Momentum in CM frame



$$\begin{aligned}
 \vec{v}_{cm} &= \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} \\
 \vec{v}_1' &= \vec{v}_1 - \vec{v}_{cm} & \Rightarrow \vec{p}_1' = m_1 \vec{v}_1' \\
 &= \vec{v}_1 - \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} \\
 &= \frac{m_2}{m_1 + m_2} (\vec{v}_1 - \vec{v}_2) \\
 \vec{v}_2' &= \frac{m_1}{m_1 + m_2} (\vec{v}_2 - \vec{v}_1)
 \end{aligned}$$

Total momentum in CM frame:

$$\begin{aligned}
 \vec{P}' &\equiv \sum_i \vec{p}_i' \\
 &= m_1 \cdot \frac{m_2}{m_1 + m_2} (\vec{v}_1 - \vec{v}_2) + m_2 \cdot \frac{m_1}{m_1 + m_2} (\vec{v}_2 - \vec{v}_1) \\
 &= 0
 \end{aligned}$$

Note: also holds for many particle system

- Elastic collision in CM frame.



in CM:

$$\vec{v}'_{1i} = \frac{m_2}{m_1 + m_2} (\vec{v}_{1i} - \vec{v}_{2i})$$

$$\vec{v}'_{2i} = \frac{m_1}{m_1 + m_2} (\vec{v}_{2i} - \vec{v}_{1i})$$

Momentum conservation:

$$m_1 \vec{v}'_{1f} + m_2 \vec{v}'_{2f} = 0$$

Energy conservation (elastic):

$$\frac{1}{2} m_1 (v'_{1f})^2 + \frac{1}{2} m_2 (v'_{2f})^2 = \frac{1}{2} m_1 (v_{1i})^2 + \frac{1}{2} m_2 (v_{2i})^2$$

$$\Rightarrow \begin{cases} \vec{v}'_{1f} = \vec{v}'_{1i} \\ \vec{v}'_{2f} = \vec{v}'_{2i} \end{cases} \quad \begin{cases} \vec{v}'_{1f} = -\vec{v}'_{1i} \\ \vec{v}'_{2f} = -\vec{v}'_{2i} \end{cases}$$

Note:

$$\vec{P}_{tot} = M \vec{v}_{cm} \text{ conserved}$$

$$E_K = \frac{1}{2} M v_{cm}^2 \text{ conserved}$$

- $D > 1$

original frame:



CM frame:

